RV College of Engineering®
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DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Semester)

Date	**04-11-2024**	**Time**	**10:00 AM-11:30 AM**	
Test	**Test-I**	**Maximum Marks**	**50**	
Course Title	**Fundamentals of Linear Algebra, Calculus & Differential Equations**		**Course Code**	**MA211TB**
Semester	**I**	**Programs**	**AS, CH, IM and ME**	

Sl. No.	**TEST-I**	M	CO	BTL
1(a)	Reduce the matrix to row echelon form and find the value of α such that rank of A is 2. $A=\begin{bmatrix}2&3&1&5&4\\0&2&2&3&4\\4&8&4&13&\alpha\end{bmatrix}$.	4	1	2
1(b)	The system of linear equations is used to analyze flow in a traffic network. The variables x, y and z represent the flow of the traffic between the three intersections in the network. Use Gauss–Seidel method to compute x, y and z with initial approximations $(0,0,0)$ for the following system of equations. $8x-3y+2z=20$, $6x+3y+12z=35$, $4x+11y-z=33$. Carry out four iterations.	6	3	3
2	Idealized spring –mass systems have numerous applications in engineering. At equilibrium, force balance equations are developed by defining the interrelationships between the springs as $2x_1+2x_2+x_3=7$, $-x_1+2x_2=-2$, $-x_1+2x_4=0$, $-3x_1+x_2+2x_3=-3$. Compute x_1, x_2, x_3, x_4 using Gauss elimination method.	10	2	2
3	Investigate the values of a and b for which the system of equations $x+2y+3z=6$, $x+3y+5z=9$, $2x+5y+az=b$, has a (i) Unique solution (ii) Infinite solution (iii) No solution. Find the solution in the case (i) and (ii).	10	1	4
4	The temperatures x_1, x_2 and x_3 of a circular metal plate observed under the same circumstances satisfy the following system of equations: $2x_1+x_2+2x_3=10$, $2x_1+2x_2+x_3=9$, $x_1+2x_2+2x_3=11$. Find the inverse for the coefficient matrix of the system using Gauss - Jordan method. Hence calculate the values for x_1, x_2 and x_3.	10	3	3
5(a)	Eigenvalues are used to determine the theoretical limit to how much information can be transmitted through a communication medium like telephone line or through the air. This is done by calculating the eigenvectors and eigenvalues of the communication channel and it is given by $\begin{bmatrix}10&2&1\\2&10&1\\2&1&10\end{bmatrix}$ with $X^{(0)}=\begin{bmatrix}1\\0\\1\end{bmatrix}$. Hence obtain the numerically largest eigenvalue and corresponding eigenvector by power method. Carry out 5 iterations.	6	4	3
5(b)	Dynamic properties of civil structures are studied through eigenvalues. For one such system represented by the following matrix $\begin{bmatrix}4&3\\2&9\end{bmatrix}$. Compute eigenvalues and corresponding eigenvectors	4	2	2

BTL-Blooms Taxonomy Level, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	-14	14	16	6	-	18	22	10	-	-